DynamiClinic

From a scoping review of 81 multi-agent studies to a system that switches its architecture per clinical case.



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github.com/LaansDole/multi-agent-simulation
branch threejs-devall

Starting point: how do *teams of LLM agents* work in healthcare?

Question

How are systems of **two or more LLM agents** that critique, debate or reach consensus **built and evaluated** in healthcare?

`Where` `Which architecture` `Which models` `How evaluated` `At what cost`

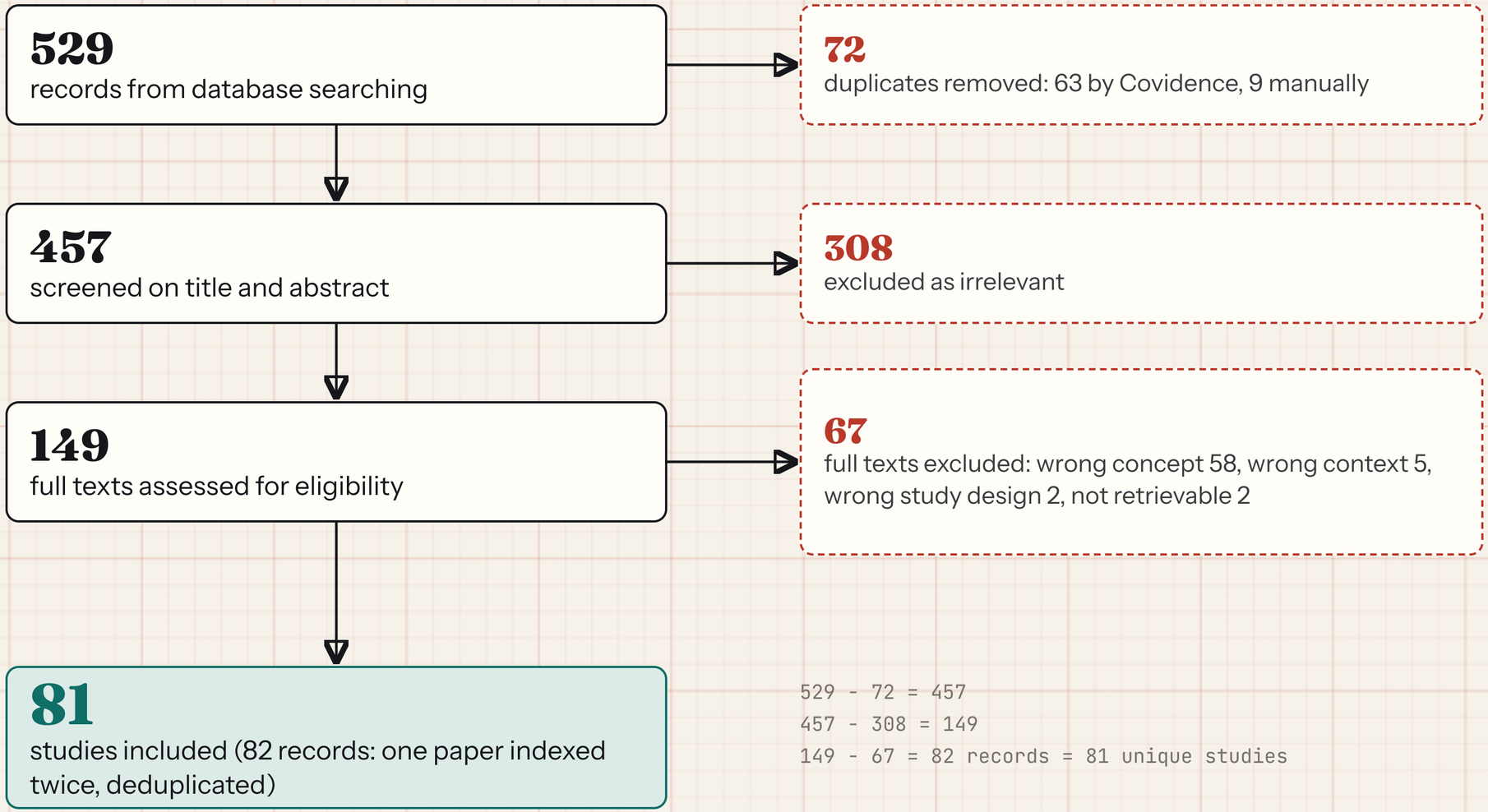
Inclusion rule

Genuine inter-agent conditioning: one agent's output is revised in response to another's. Not single-LLM tools, not fixed role chains without feedback.

Method

- **5 databases**: PubMed, Scopus, CINAHL, Cochrane Library, IEEE Xplore
- Searched June 2026, records from 2022 to 2026
- **Arksey and O'Malley** five stages, reported per **PRISMA-ScR**
- Screening and full-text review in **Covidence**
- Every charted value backed by a **verbatim quote** from the source

529 records in, *81* studies out




```
529 - 72 = 457
457 - 308 = 149
149 - 67 = 82 records = 81 unique studies
```

PRISMA-ScR flow diagram, Fig. 1 of the scoping review

Architecture is the *design variable*; cost and safety lag behind

6

architecture families

Debate and consensus 35, hierarchical coordinators 30, dynamic team assembly only 11. Families overlap.

4-12x

the cost of collaboration

4-12x longer than single-modality tools (Wan 2025); 3.6x the tokens and time of baselines (Mandal 2025).

between agents

where failure lives

Collusion, unsafe content spreading across agents, and false consensus that silences a correct minority.

7 / 81

tested with people

Benchmarks (35) and retrospective data (22) dominate; no randomized trial. Call: compare against strong single-model baselines.

Scoping review, sections 3 and 4. Counts are per study and overlap.

Each finding became a *design requirement*

REVIEW FINDING	REQUIREMENT	DYNAMICLINIC
Only 11 of 81 assemble the team per case; MDAgents picks solo or team at the start	Choose the architecture per case	**Meta-Controller with three modes**
Collaboration takes 4-12x longer; 3.6x the tokens	Spend the team only where needed	**Hierarchical: one agent for simple, certain cases**
False consensus and collusion between agents	Engineer disagreement and checks	**Adversarial debate + safety gates**
Few head-to-head comparisons with a single model	Always measure against one agent	**Three arms: single, static team, dynamic**
Interaction transcripts are the multi-agent model cards	Make every hand-off inspectable	**Shared Blackboard + 3D office**

Findings: scoping review sections 3-4 (future directions 4.3). Requirements and design: DynaClinic research proposal.

A Meta-Controller makes the *topology a variable*

Two scores, one rule

Complexity C: how dense the case is (comorbidities, drugs, conflicting labs).
Uncertainty U: entropy of a generalist's first differential.

```
U > tau_u → adversarial
C ⩽ tau_c → hierarchical
otherwise → collaborative
```

A Conflict Detector escalates collaborative to adversarial when the specialists diverge. tau_u is calibrated per model (gpt-6-luna: 0.15 multiple choice, 0.59 open-ended).

Hierarchical
SPEED

Primary Care Agent as the hub; specialists only on request.

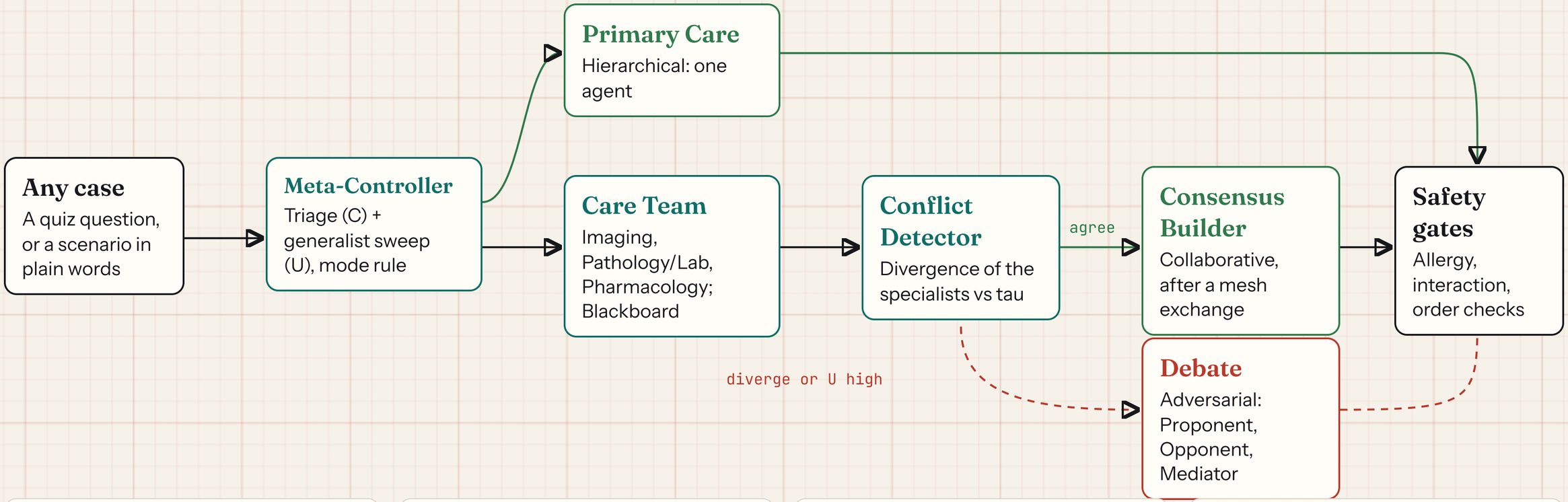
Collaborative
BREADTH

Imaging, Pathology/Lab, Pharmacology on a shared Blackboard; mesh exchange; Consensus Builder.

Adversarial
DEPTH

Proponent defends H1, Opponent (Clinical Critic) attacks, Mediator decides.

Built as one workflow on the *DevAll* multi-agent platform



10
custom node types in one YAML workflow

3
evaluation arms: single, static team, dynamic

3D
office: every agent a character showing its live role

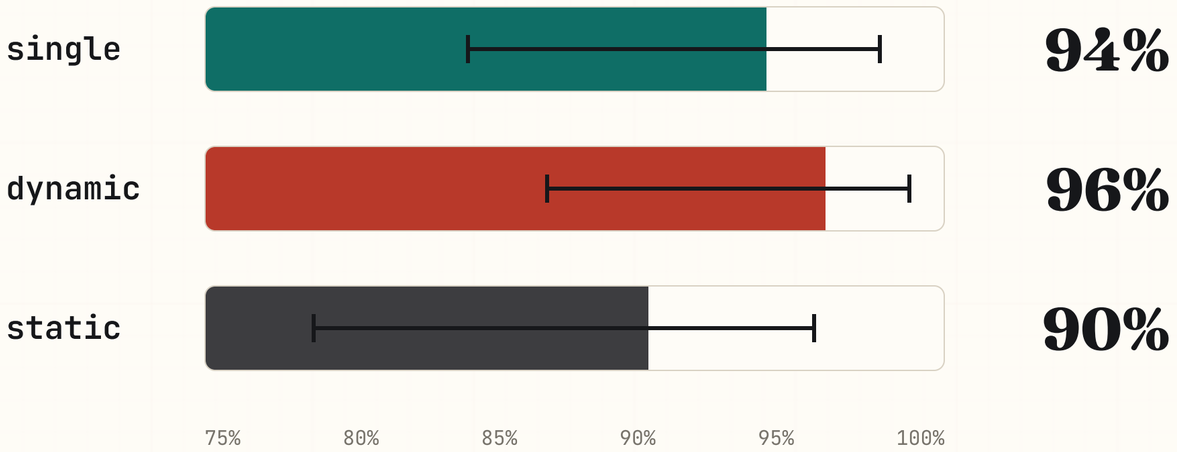
1,030
automated tests, no model calls needed

yaml_instance/dynaclinic_mvp.yaml / dynaclinic/ / the-delegation/ (React + Three.js). Stubs: FHIR/Neo4j, MCTS planner, BioBERT/Qdrant memory.

Works end to end; *not yet better* than one agent

MedQA-USMLE, 50 test questions, gpt-6-luna

Accuracy with 95% interval, one run per arm.



	SINGLE	DYNAMIC	STATIC
gpt-6-luna calls / question	1.00	6.94	6.00
gpt-6-luna mean latency	5.3 s	32.0 s	29.9 s
qwen3.5-4b accuracy (local)	68%	60%	62%

The team's answer follows the Meta-Controller's **first look** (gpt-6-luna 47 of 50, qwen 31 of 50); on exam questions the specialists add little. All intervals overlap: exactly the **single-model comparison** the review asks for.

MedQA: Jin et al. 2021, 4-option test split, seed 2026 sample. qwen: dynamic with its own calibrated tau_u; single vs dynamic p = 0.34.

Next: make the team *earn its cost*

1 **Harder cases, mechanism by mechanism**

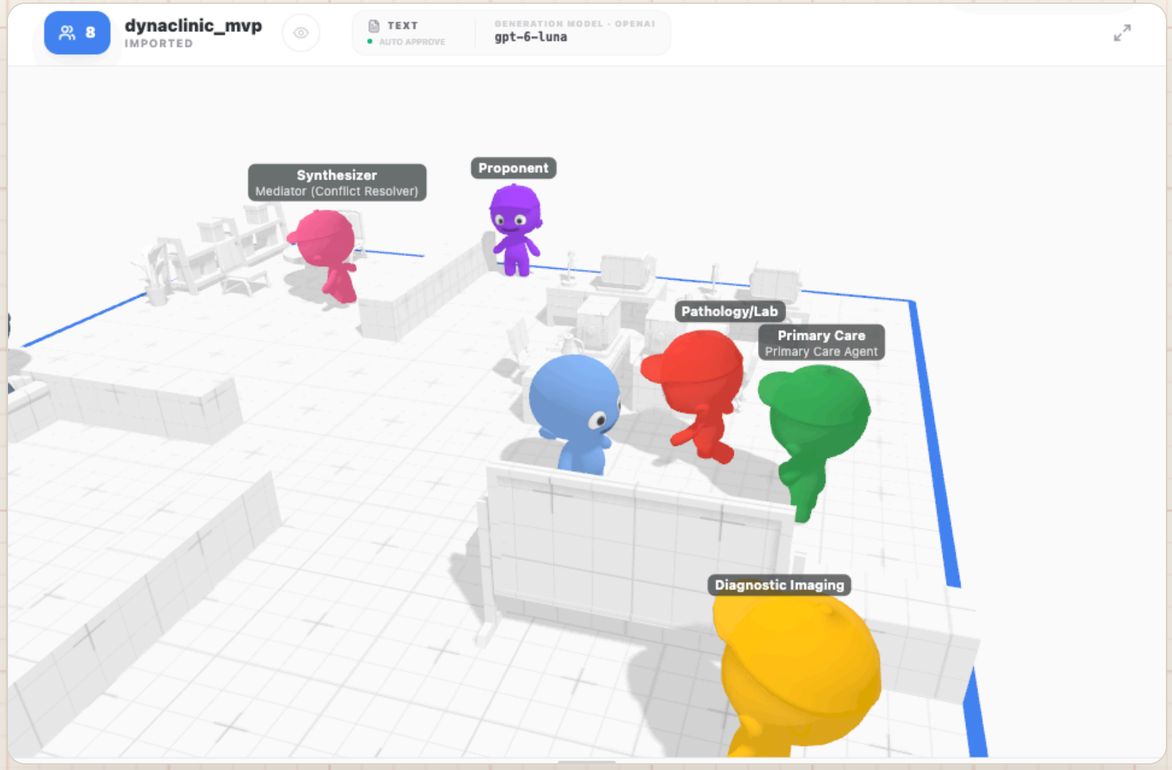
A benchmark where one agent fails (MedXpertQA); ablate mesh, debate and gates one at a time.

2 **A better controller**

Validate tau against outcomes; give the first look a vote; a Mediator that weighs evidence, not plurality.

3 **Toward clinical validation**

Clinician review of debate transcripts, then the prospective studies the review found missing.



LIVE DEMO: ONE CASE PER MODE

A	Cellulitis, sulfonamide allergy	HIERARCHICAL
B	Renal colic on phenelzine	COLLABORATIVE
C	30 weeks pregnant, chest pain	ADVERSARIAL